

# Roof Lifecycle Cost Analysis

Planned maintenance is projected to save the owner of a typical Hampton Roads medical facility **\$729K** over the next 20 years.

About 37% less than reactive repair: \$553K lower spending plus \$176K more roof life at year 20. Range \$382K–\$947K from conservative to upside assumptions; \$376K in today's dollars at a 7% discount rate.

| MARKET        | PROPERTY | ROOF AREA    | SYSTEM           | AGE · CONDITION | NEW ROOF TODAY |
|---------------|----------|--------------|------------------|-----------------|----------------|
| Hampton Roads | Medical  | 60,000 sq ft | Modified bitumen | 8 yrs · Good    | \$480K         |

**PAXSEAL NET COST**

**\$1.25M**

\$1.21M spent + \$288K roof used – \$240K roof left

**REACTIVE NET COST**

**\$1.98M**

\$1.76M spent + \$288K roof used – \$64K roof left

**REPAIR & STORM SAVINGS**

**\$884K**

Unbudgeted repairs and storm damage avoided

**ROOF PRICES**

**2× in 15 yrs**

At 4.6%/yr, the long-run trend in commercial roofing costs

## Cumulative roof spending, next 20 years

Nominal dollars. Vertical jumps are full roof replacements.



## Roof life

When each scenario needs a new roof.



Where the savings come from

20-year totals

|                           | PAXSEAL | REACTIVE | SAVINGS |
|---------------------------|---------|----------|---------|
| PaxSeal fees              | \$231K  | —        | -\$231K |
| Repairs paid by owner     | \$117K  | \$966K   | +\$849K |
| Expected storm damage     | \$70K   | \$105K   | +\$35K  |
| Roof replacements         | \$787K  | \$688K   | -\$99K  |
| Cash spending             | \$1.21M | \$1.76M  | +\$550K |
| Current roof value used   | \$288K  | \$288K   | —       |
| Roof life left at year 20 | -\$240K | -\$64K   | +\$176K |
| Net 20-year cost          | \$1.25M | \$1.98M  | +\$730K |

What this means

- 1 A later replacement. The next new roof moves from year 8 to year 11, leaving 10 years of roof life at year 20 instead of 2.
- 2 Smaller repairs. Emergency repairs run about 4.2× the planned rate in a medical facility, and the PaxSeal fee covers half of planned repairs. Together that saves \$849K.
- 3 Patient care keeps running. A leak in a clinical area triggers infection-control work, closes rooms and can damage equipment.
- 4 The program pays for itself. Each \$1 of fees avoids about \$3.39 of other cash roof costs.

Local risk profile

Norfolk city, VA (51710)

Hurricane / tropical-storm wind

About 4.8 tropical storms or stronger within 50 nautical miles of Norfolk per 20 years (NOAA HURDAT2 1975–2024). Expected storm damage to this roof over 20 years is \$70K if maintained and \$105K with reactive repair only, before insurance. A documented maintenance history also helps when an insurer reviews a claim.

|                                 |                     |       |
|---------------------------------|---------------------|-------|
| Hail                            | Relatively Low      | ■■■■■ |
| Strong wind                     | Relatively Moderate | ■■■■■ |
| Tornado                         | Relatively Moderate | ■■■■■ |
| Hurricane                       | Relatively Moderate | ■■■■■ |
| Winter weather                  | Relatively Moderate | ■■■■■ |
| Ice storm                       | Relatively Moderate | ■■■■■ |
| FEMA National Risk Index rating |                     |       |

Key assumptions

The basis for each figure in this analysis.

| ASSUMPTION           | VALUE            | BASIS                                                                                                   |
|----------------------|------------------|---------------------------------------------------------------------------------------------------------|
| Roof life            | 20 / 15 yrs      | Maintained / reactive. Industry life tables; maintenance adds ~5 yrs (NRC Canada, warranties)           |
| Roof cost growth     | 4.6%/yr          | BLS Producer Price Index, nonresidential roofing, 2007–2025                                             |
| Replacement cost     | \$8.00/sq ft     | Hampton Roads contractor price ranges                                                                   |
| Reactive repair cost | 3× planned × 1.4 | Industry estimates; ×1.4 for leak damage (medical facility). PaxSeal fee covers half of planned repairs |
| Storm damage         | 0.44%/yr         | Of roof cost, before insurance; ×1.5 if reactive. FEMA National Risk Index                              |
| PaxSeal fee          | \$8,600/yr       | Year 1, rising 3%/yr. Estimate; your proposal states the actual fee                                     |

**Next step:** a roof assessment and a written PaxSeal proposal for this building. Call PAX Services Group at 1-855-729-7510 or email [info@paxserv.com](mailto:info@paxserv.com).

**Estimates, not a quote.** Projections from Hampton Roads market research and industry data, in nominal dollars. The PaxSeal fee is an estimate; your proposal states the actual price. Net cost includes wear on the current roof, less roof life left at year 20; both roof-value figures are in today's dollars. Storm damage is before insurance.